



Search and Rescue drone

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Problem Statement

How to design a drone swarm capable of effectively locating, detecting, and alerting to the presence of a person in distress?



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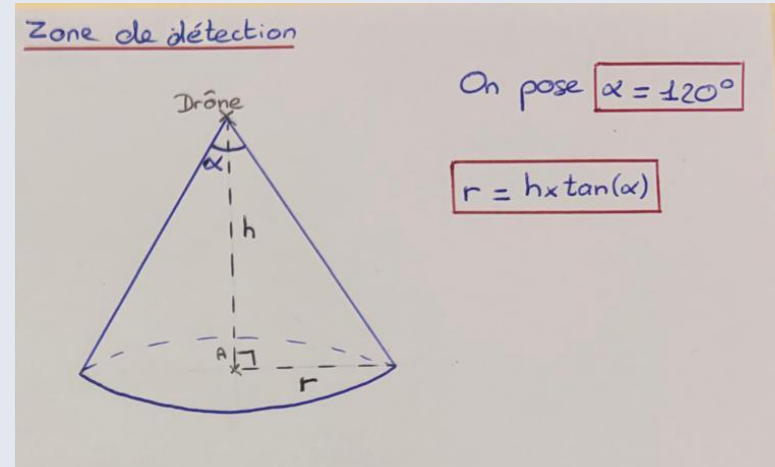
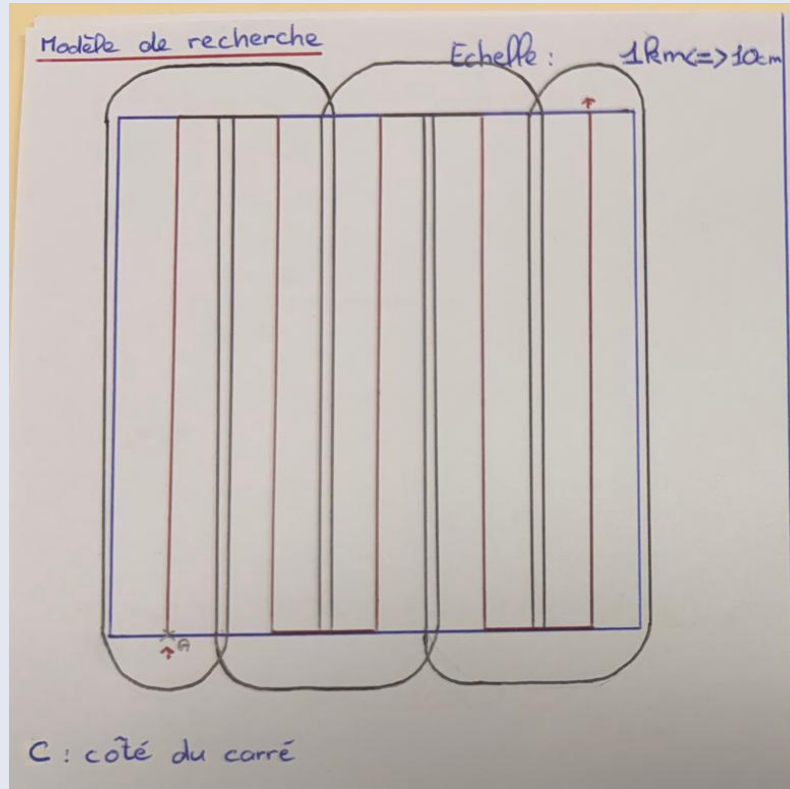
Gantt Chart





01 Human Search

Search for individuals on a flat 1 km² plot



Search for individuals on a flat 1 km² plot

$$Na = \frac{c}{(2 * r * 95\%)}$$

$$D = Na * c + (Na - 1) * 2 * r * 95\%$$

$$A = \frac{D}{v}$$

$$Nd = \left\lceil \frac{A}{a} \right\rceil + 1$$

r: radius of the base of the cone

Na: number of passes required to cover (grid) the area

D: total distance to be traveled

A: total endurance

v: drone speed (20 km/h)

a: endurance of a drone (10 or 20 minutes)

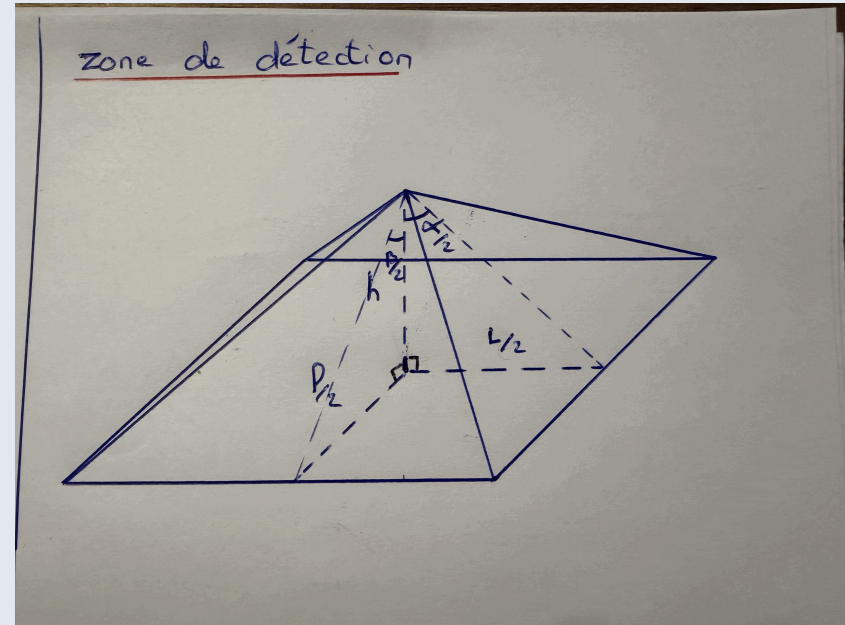
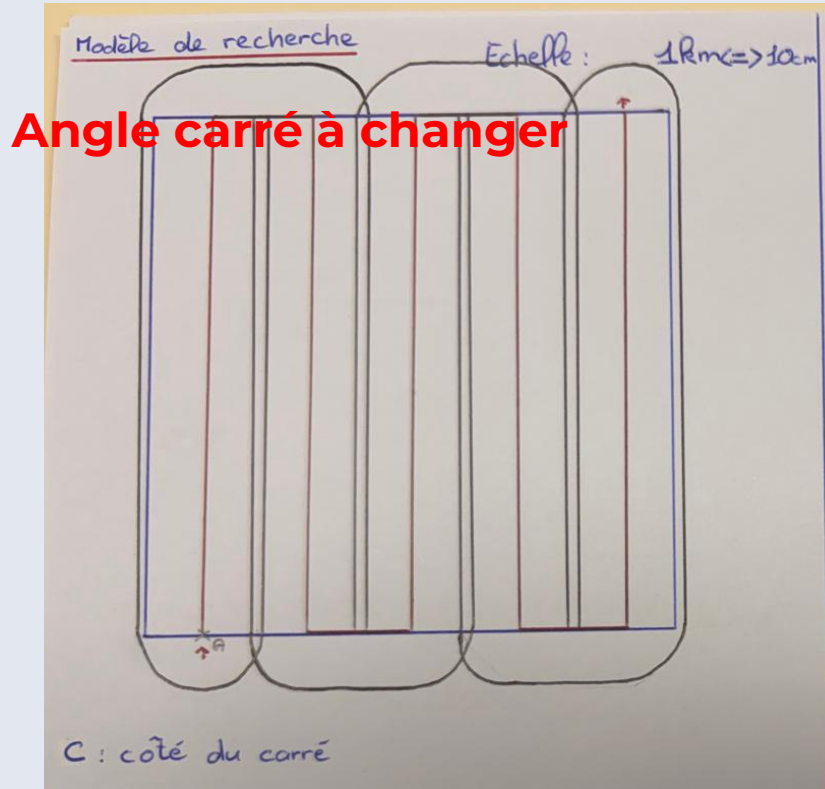
Nd: number of drones required

Search for individuals on a flat 1 km² plot

h (m)	D (m)	A (h)	Nd
2	6,93	8,1	49
4	13,86	4,1	25
6	20,78	2,7	17
8	27,71	2,1	13
10	34,64	1,7	10

a = 10 min

Search for individuals on a flat 1 km² plot



$$\alpha = 31^\circ$$

$$\beta = 23^\circ$$

Search for individuals on a flat 1 km² plot

$$L = 2 * h * \tan\left(\frac{\alpha * \pi}{2 * 180}\right)$$

$$l = 2 * h * \tan\left(\frac{\beta * \pi}{2 * 180}\right)$$

$$Na = \frac{c}{(L * 95\%)}$$

$$D = Na * c + (Na - 1) * L * 95\%$$

$$A = \frac{D}{v}$$

$$Nd = \left\lceil \frac{A}{a} \right\rceil + 1$$

L: length of the rectangular base

l: width of the rectangular base

Na: number of passes required to cover (grid) the area

D: total distance to be traveled

A: total endurance

v: drone speed (20 km/h)

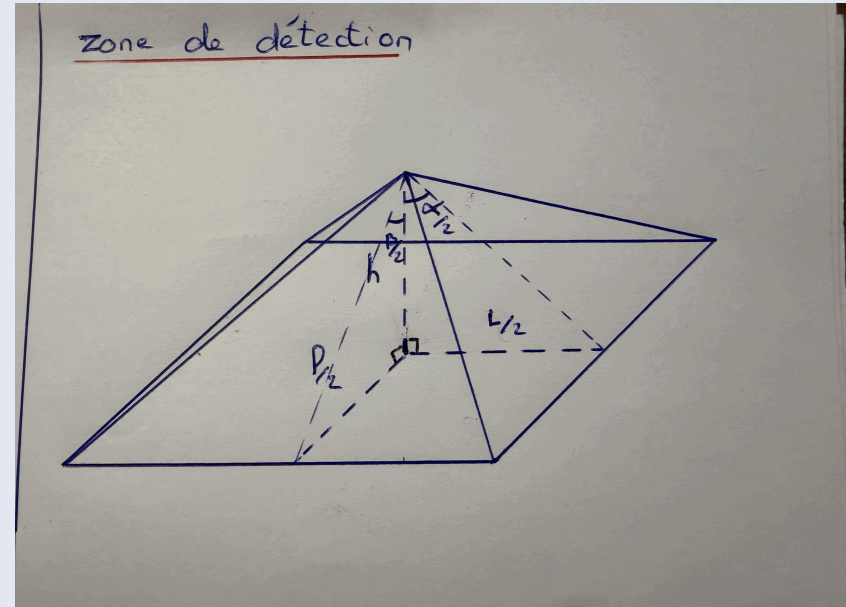
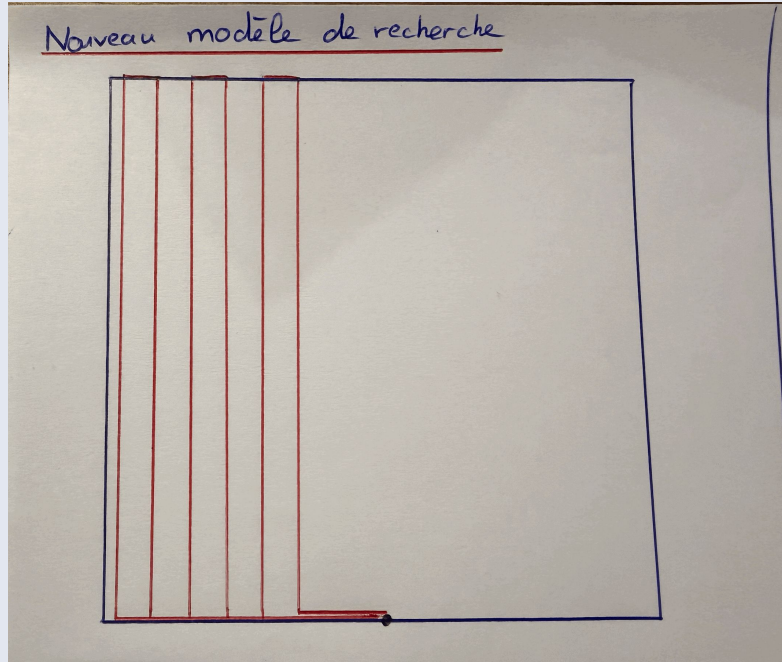
a: endurance of a drone (10 or 20 minutes)

Nd: number of drones required

Search for individuals on a flat 1 km² plot

Height (m)	Length (m)	Width (m)	Nb of passes	distance (m)	endurance (s)	endurance (min)	Nb of drones 10 min	Nb of drones 20min
2	1,11	0,81	949	949916	170985	2850	285	143
3	1,66	1,22	633	633610	114050	1901	191	96
4	2,22	1,63	474	475456	85582	1426	143	72
5	2,77	2,03	380	380564	68502	1142	115	58
6	3,33	2,44	316	317302	57114	952	96	48
7	3,88	2,85	271	272115	48981	816	82	41
8	4,44	3,26	237	238225	42880	715	72	36
9	4,99	3,66	211	211866	38136	636	64	32
10	5,55	4,07	190	190778	34340	572	58	29

Search for individuals on a flat 1 km² plot



Search for individuals on a flat 1 km² plot

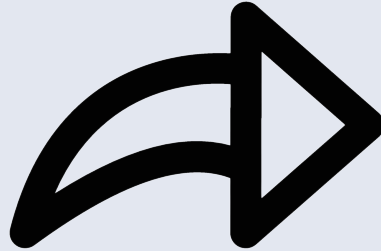
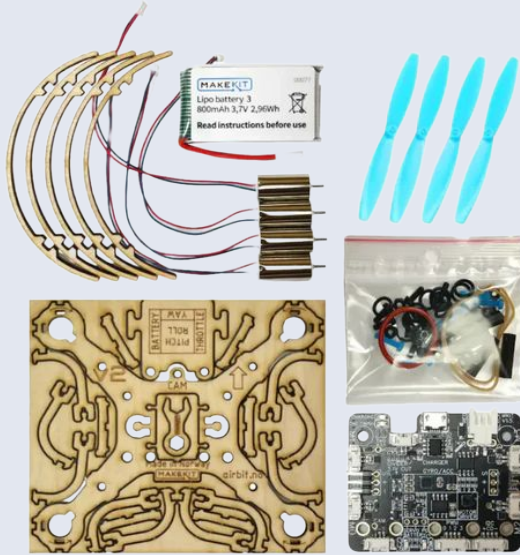
Height (m)	Length (m)	Width (m)	Nb of passes	distance (m)	distance covered by one drone (m)	Nb of drones
2	1,11	0,81	949	949916	6005	159
3	1,66	1,22	633	633610	6008	106
4	2,22	1,63	474	475456	6011	80
5	2,77	2,03	380	380564	6013	64
6	3,33	2,44	316	317302	6016	53
7	3,88	2,85	271	272115	6018	46
8	4,44	3,26	237	238225	6021	40
9	4,99	3,66	211	211866	6024	36
10	5,55	4,07	190	190778	6026	32

For H=10 : 22 min → 7,26 km

02 Drone Conception



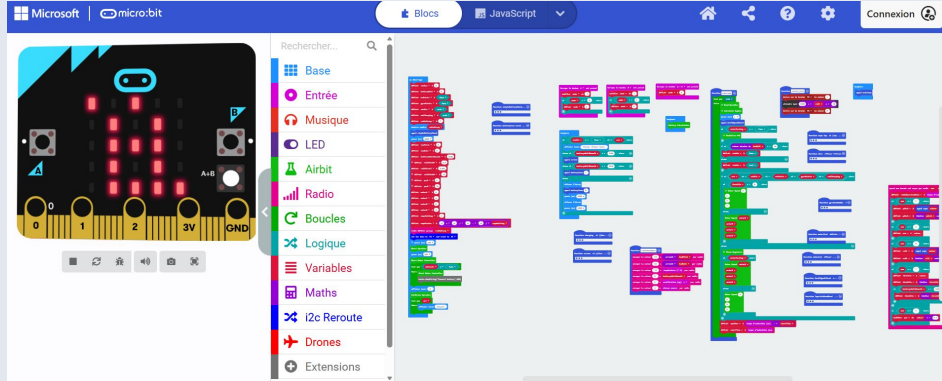
Drone Air bit 2



Specifications::

- **Flight time:** 7-10 min
- **Speed:** 20 km/h

Drone Programming

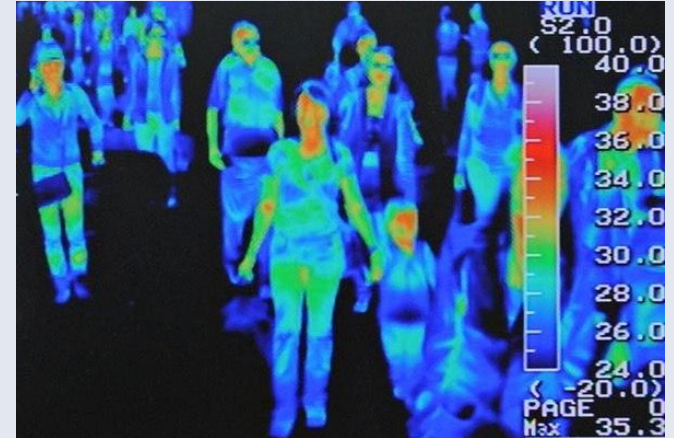


MakeCode





03 Thermal Analysis

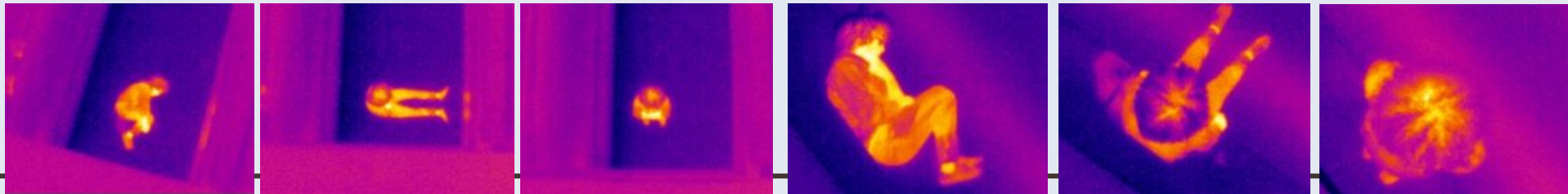


Thermal Analysis

Indoor testing,

Based on Hypothesis 1 at 21°C (School interior)

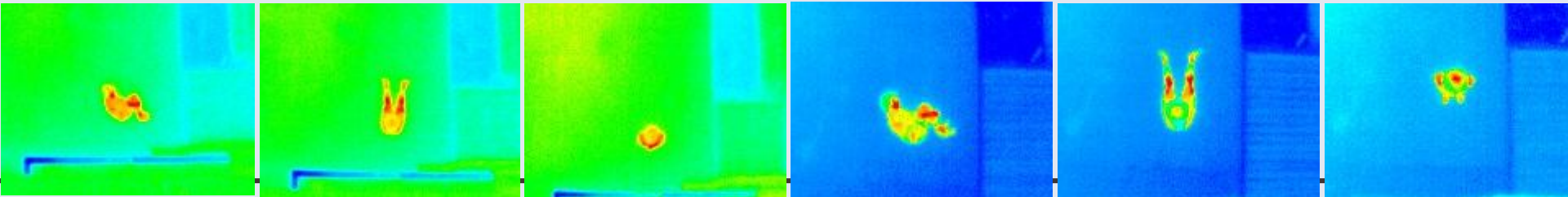
Resolution	Field of View (FoV)		
160*120 pixel	31° par 23°		
Altitude (m) / Physical postures	Standing(°)	Sitting(°)*	Curled up (°)
3	24,4	24,9	24,3
7,6	21,4	21,6	22,4
*Sitting with legs extended			



Thermal Analysis

Outdoor testing,

Based on Hypothesis 2 at 15°C (School Outdoor)			
Resolution	Field of View (FoV)		
160*120 pixel	31° par 23°		
Altitude (m) / Physical postures	Standing(°)	Sitting(°)*	Curled up (°)
7,6	14,2	14,2	14,3
10,5	12,9	12,6	12,4
*Sitting with legs extended			

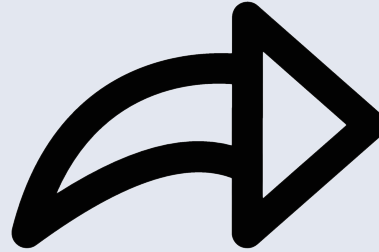
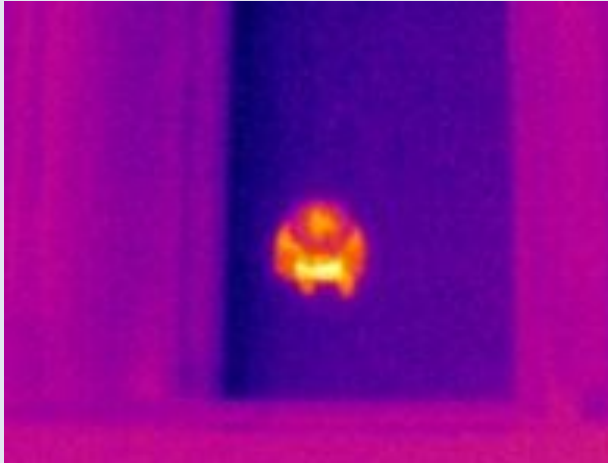


Thermal Analysis

Height (m)	Width (m)	Nb of photos/s	Nb of photos/1000m
2	0,81	7	1294
3	1,22	5	863
4	1,63	4	647
5	2,03	3	518
6	2,44	3	432
7	2,85	2	370
8	3,26	2	324
9	3,66	2	288
10	4,07	2	259

Thermal Analysis

From thermal input to data analysis



010	000
000	010
010	000

04

Image Processing

Goal and development

Main goal :

- detect the presence of a human being

Development logic

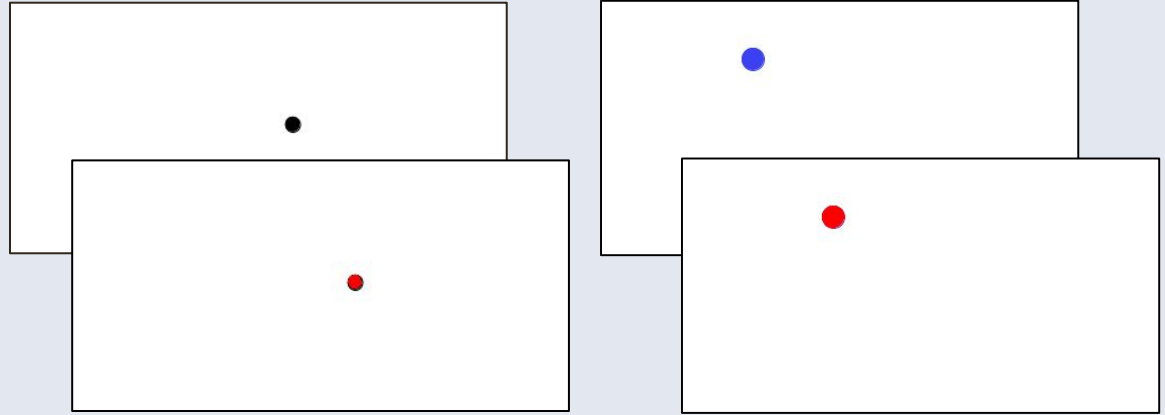
- Establish simple hypotheses
- Complicate them, step by step, to reflect the reality on the ground

Search with a normal camera



We must detect a singularity in the landscape

Code version 1 and version 2

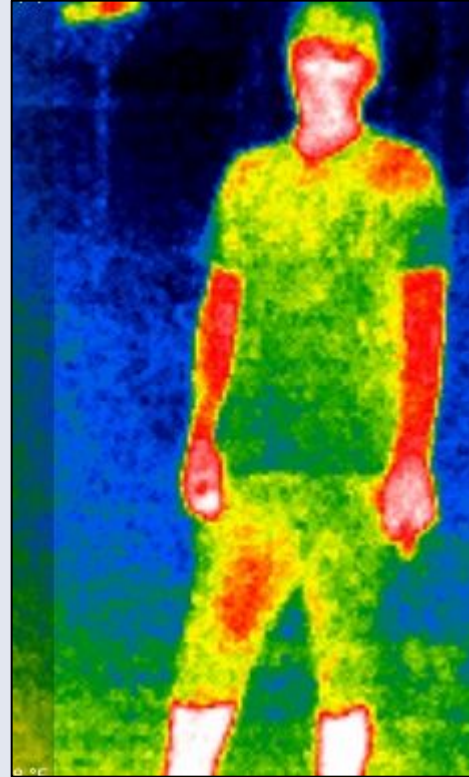


Version du code	Nom image	Postion à déterminer	Position déterminée	% d'erreur	% d'erreur total
1	Blanc_Point_Noir	205	206	0.5	0.3
		206	205.7	0.1	
		205	206	0.5	
1	Blanc_Point_Noir	206	205.7	0.1	0.3
		77	76.1	1.2	
		37	37.1	0.3	
1	Blanc_Point_Noir_2	178	178.1	0.1	0.75
		131	130.1	0.7	
		228	228.1	0	
1	Blanc_Point_Noir_4	100	100.1	0.1	0.05
		249	0	100	
		154	0	100	
1	Blanc_Point_Rouge	249	248.5	0.2	100
		153	153.9	0.6	
		51	51.2	0.4	
2	Blanc_Point_Rouge_2	60	60.1	0.2	0.3
		195	194.8	0.1	
		223	223.1	0	
2	Blanc_Point_Rouge_3	268	266.2	0.7	0.05
		108	106.2	1.7	
		108	106.2	1.7	

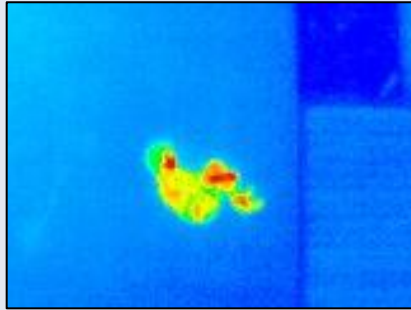
Search with a thermal camera

Why use a thermal imaging camera?

- Climate difficulty
- Night-time is a challenge



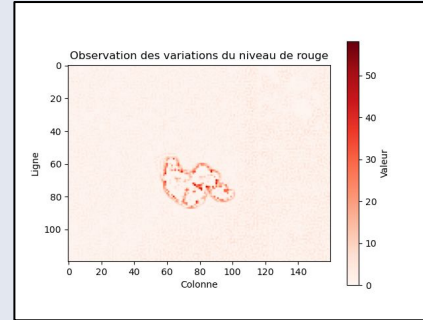
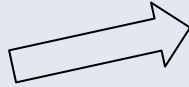
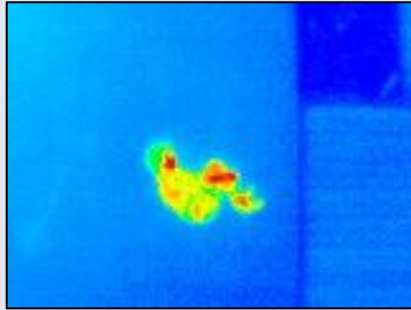
Goal and development



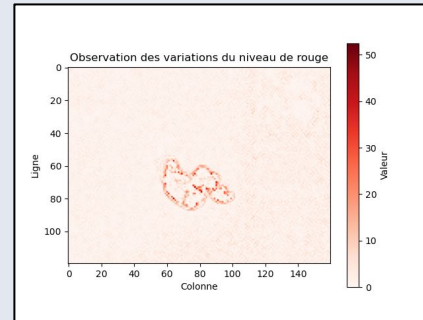
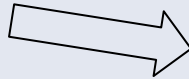
Paradigm :

- Significant variation in the red spectrum (RGB) in the presence of a human being
- Rest of the landscape : monochrome

Thermal search code

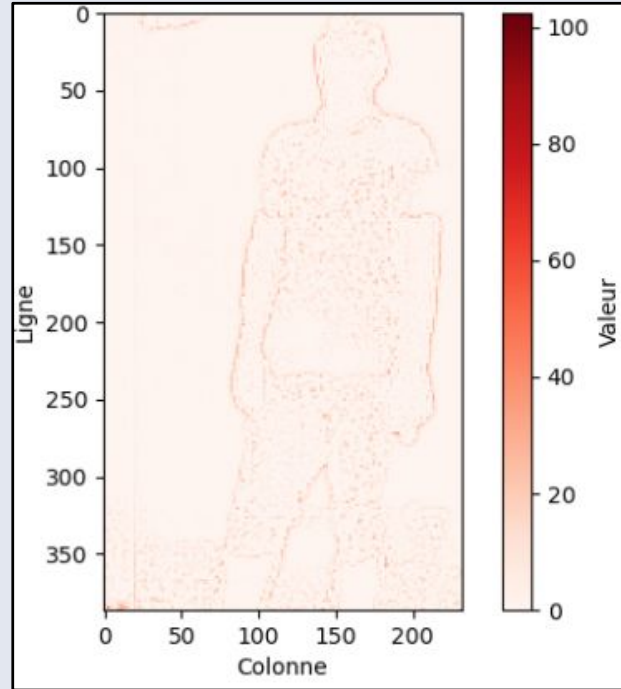
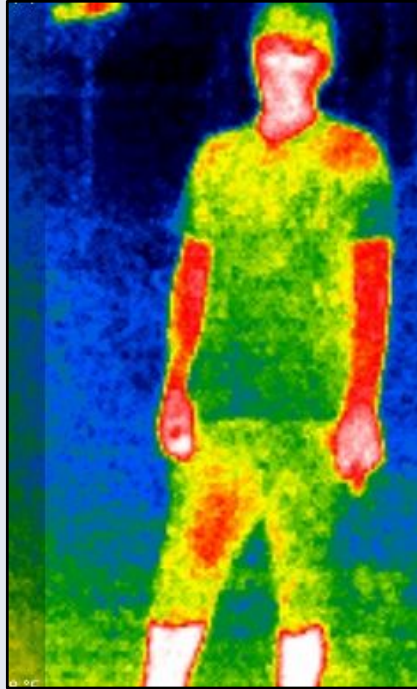


Calcul de variations
avec les 8 voisins



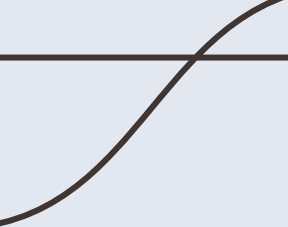
Calcul de variations
avec les 4 voisins

The outline of a human being



Conclusion

- Increasingly precise research to define the detection zone and how to detect a person
- Connecting drones to each other
- Feedback to base
- Further reducing the number of drones
- Program adaptability in different environments
- Determining the characteristics of a human being. ML for training a detection programme



Thank you for your attention.

Ilona BERTIN, Mélina BRUTIER, Louis OSCABY, Florent NASSCHAERT